Sheshadri Rathan C S

es24btech11030@iith.ac.in | sheshadrirathan77@gmail.com | +91 9108173865 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.10
SRI EDU PU COLLEGE	Class XII (KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD)	2024	96.50%
NEWTON GRAMMAR HIGH SCHOOL	Class X (KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD)	2022	99.20%

SCHOLASTIC ACHIEVEMENTS

- Qualified for the **NASA/NSS International Space Development Conference (ISDC) 2018**, where the team's space settlement project was selected for scientific poster presentation.
- Secured an SGPA of **9.67** in the 2nd semester, ranking as one of the highest scores across the entire engineering branch.
- Ranked in the **top 3** of the academic cohort, securing selective admission into the **Computer Science specialization.**

SKILLS

- **Programming Languages:** C, C++, Python, HTML, CSS
- **Databases:** PostgreSQL
- **Core Computer Science:** Data Structures and Algorithms, Object-Oriented Programming (OOP), STL
- **Tools & Technologies:** Git, GitHub, NumPy

RELEVANT COURSES

- **Core Courses:** Introduction to Programming, Algorithms, Data Structures and Applications, Introduction to Database Management Systems, Theory of Computation, Discrete Mathematics for Computer Science, Operating Systems-1
- **Mathematics Courses:** Calculus I, Calculus II, Probability and Random Variables, Introduction to Statistics, Elementary Linear Algebra, Differential Equations, Complex Variables

PROJECTS

Movie Ticket Booking System (C++) | *C++, OOP, STL, File Handling, Git*

- Developed a console-based **Movie Ticket Booking System** using **C++**, **Object-Oriented Programming (OOP)**, **STL**, and **file handling** to manage movies, shows, and customer bookings.
- Implemented **ticket booking, seat allocation, booking history**, and persistent text-file storage to ensure efficient and reliable data management.
- Designed a modular architecture with dedicated **Movie**, **Show**, **Booking**, and **BookingSystem** classes, enabling clear separation of responsibilities, improved maintainability, and scalability.

Rare-Event Probability Estimator (Python) | *Python, NumPy, Monte Carlo Simulation, Importance Sampling*

- Developed a **Rare-Event Probability Estimator** using **Python**, **NumPy**, and Monte Carlo simulation to estimate extremely low-probability events in the tails of the standard normal distribution.

- Implemented **Importance Sampling** with a shifted proposal distribution and a custom **Box-Muller transform**, generating over **10 Gaussian samples** while eliminating dependence on built-in normal random number generators.
- Achieved **multiple orders of magnitude reduction in estimator variance** compared to standard Monte Carlo methods, significantly improving the accuracy and computational efficiency of rare-event probability estimation.

Producer-Consumer Synchronization |*C, System V IPC, Shared Memory, Semaphores, Linux, Git*

- Implemented a bounded buffer **Producer-Consumer** system using **System V Shared Memory** and **three semaphores** for synchronized inter-process communication.
- Designed a shared circular buffer with **atomic multi-semaphore operations (semop())** to coordinate concurrent producer and consumer processes while preventing race conditions.
- Analyzed process synchronization through synchronized execution logs, demonstrating practical concepts of mutual exclusion, resource sharing, and concurrent process coordination.

Coding Tracker (C++) | *C++, OOP, STL, File Handling, Git*

- Developed a console-based **Coding Tracker** using **C++**, **Object-Oriented Programming (OOP)**, **STL**, and **file handling** to efficiently organize and persist coding practice records.
- Implemented features for **adding, updating, deleting, searching, and tracking** coding problems with persistent text-file storage, enabling efficient record management.
- Built a modular architecture with dedicated **Problem** and **Tracker** classes, along with a statistics module to analyze solved problems, revision counts, and topic-wise distribution for monitoring coding progress.

POSITIONS OF RESPONSIBILITY

Speaker Curator & Content Coordinator, EML (Extra Mural Lectures), IIT Hyderabad

- Coordinated speaker outreach by identifying, contacting, and inviting distinguished professionals for lecture sessions.
- Assisted in planning and executing lecture events, managing coordination and on-ground event logistics.
- Designed promotional content for social media to increase event visibility and audience engagement.

EXTRACURRICULAR

- Secured 2nd Place in an **E-Waste Management Ideathon** for proposing an innovative solution to sustainable e-waste management.
- Regularly solve DSA problems on LeetCode to strengthen problem-solving and coding skills.
- Volunteered with the **National Service Scheme (NSS)** at IIT Hyderabad, contributing to community service and social outreach initiatives.